

CORNELL UNIVERSITY

AI360 Certificate Program

Finding Patterns in Data Using Cluster and Hotspot Analysis

Geospatial Hotspot Analysis of COVID-19 Vaccine Penetration Across the Continental United States

Getis-Ord Gi Local Statistic | County-Level Analysis | September 1, 2021*

Course Project, Part Two

March 2026

Dataset: CDC Vaccination Data + Continental US County Shapefile

3,000+ Counties | Polygon-Based Spatial Weights | Getis-Ord Gi*
Statistic

Tools: R | spdep | ggplot2 | sf

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1. Executive Summary

This report presents a comprehensive geospatial hotspot analysis of COVID-19 vaccine penetration across all counties of the continental United States as of September 1, 2021. The analysis was completed as Course Project, Part Two, for the course Finding Patterns in Data Using Cluster and Hotspot Analysis within the AI360 Certificate Program at Cornell University. Using the Getis-Ord G_i^* local statistic implemented in R through the `spdep` package, county-level vaccination rates were evaluated for statistically significant spatial clustering, identifying regions where high and low vaccination rates are geographically concentrated beyond what would be expected by chance.

The analysis reveals a striking and geographically coherent pattern of spatial inequality in vaccine uptake. The Deep South — including Alabama, Mississippi, Georgia, Tennessee, and Arkansas — emerges as a dominant coldspot region with G_i^* z-scores reaching as low as -6.67, indicating vaccination rates clustered seven standard deviations below what would be expected under spatial randomness. Conversely, the Northeast corridor and New England represent strong hotspot clusters, with maximum z-scores reaching +5.18. The majority of central US counties fall within a statistically insignificant range, with an interquartile range from -1.05 to +1.21.

Key Findings at a Glance

- G_i^* z-scores range from -6.67 (strongest coldspot) to +5.18 (strongest hotspot)
- Deep South is the dominant coldspot: Alabama, Mississippi, Georgia, Tennessee, Arkansas
- Northeast and New England form the dominant hotspot region with z-scores above +4
- Median z-score of -0.0006 and mean of +0.000144 confirm proper centering of the statistic
- Only 2 counties returned NA values due to empty neighbor lists (handled via `zero.policy=TRUE`)
- Texas data is limited (sourced from Texas Tribune) — results in that region warrant caution
- Tools used: R, `spdep`, `ggplot2`, `sf` packages

2. Introduction and Background

2.1 Research Context

The COVID-19 pandemic produced one of the most geographically uneven public health responses in modern American history. While vaccines became widely available in the United States by spring 2021, uptake varied dramatically across regions, states, and individual counties. By September 2021, a clear geographic divide had emerged: highly vaccinated urban and northeastern communities contrasted sharply with low-uptake rural and southern communities. Understanding the spatial structure of this divide — and specifically identifying statistically significant hotspots and coldspots of vaccine penetration — is critical for designing effective public health interventions.

Hotspot analysis using the Getis-Ord G_i^* statistic is particularly well suited for this type of question. Unlike simple mapping of raw vaccination rates, the G_i^* statistic tests whether the spatial clustering of high or low values is statistically significant given the spatial arrangement of neighboring counties. This provides a rigorous, probabilistic basis for identifying truly clustered regions rather than isolated high or low observations, making it a powerful tool for public health geography.

2.2 Course Context

This project was completed as part of the Cornell University AI360 Certificate Program course Finding Patterns in Data Using Cluster and Hotspot Analysis. The AI360 program provides a rigorous foundation in artificial intelligence and data science methods with emphasis on real-world application. This project specifically demonstrates skills in spatial data processing, geospatial analysis, and the application of local spatial statistics using R — competencies directly relevant to roles in data science, public health analytics, urban planning, and policy analysis.

2.3 Research Objectives

1. Load and join county-level geographic and vaccination data using FIPS codes as the common identifier.
2. Construct a spatial weights matrix based on polygon adjacency using the `poly2nb()` and `nb2listw()` functions.
3. Compute the Getis-Ord G_i^* local statistic for each county using the `localG()` function.
4. Visualize the resulting z-scores as a choropleth map of the continental United States.
5. Interpret the spatial patterns of vaccine penetration clustering and derive public health insights.
6. Evaluate methodological choices, including the use of centroid vs. polygon-based neighborhood definition and the role of spatial weighting.

3. Data Sources and Description

3.1 Vaccination Data

The primary data source is VaxCountySept012021.csv, which contains CDC-reported COVID-19 vaccination statistics for US counties as of September 1, 2021. The dataset includes multiple vaccination metrics by age group and dose completion status. The primary variable of interest for this analysis is Series_Complete_Pop_Pct — the percentage of the total county population that had completed the full vaccine series (two doses of Moderna or Pfizer, or one dose of Johnson & Johnson) as of the analysis date. An important data caveat is that Texas county data was sourced from the Texas Tribune website rather than the CDC, due to incomplete CDC reporting for that state, and should be interpreted with appropriate caution.

3.2 Geographic Data

The geographic data consists of two shapefiles. The continentalUSt.shp file provides county-level polygon boundaries for all counties in the continental United States, covering approximately 3,000 counties across 48 states. The Counties.shp file provides New York State county boundaries used in the companion analysis script. Both shapefiles were loaded using the st_read() function from the sf package in R, which reads and processes spatial data in standard shapefile format.

3.3 Data Join

The vaccination data was joined to the county shapefile using the combine column, which contains the concatenated State FIPS code and County FIPS code for each observation (e.g., state FIPS 1 + county FIPS 001 = combine value 1001 for Autauga County, Alabama). This FIPS-based join is the standard approach for linking census-geographic and administrative datasets in the United States and ensures that each vaccination record is correctly matched to its corresponding county polygon.

Variable	Description	Type
combine	Concatenated State + County FIPS code (join key)	Integer
Series_Complete_Pop_Pct	% of total population fully vaccinated	Numeric (%)
Series_Complete_Yes	Count of fully vaccinated individuals	Integer
Administered_Dose1_Pop_Pct	% of population with at least one dose	Numeric (%)
Series_Complete_65Plus	Fully vaccinated count, age 65+	Integer
SVI_CTGY	Social Vulnerability Index category	Categorical
NAME	County name	Character
Recip_State	State abbreviation	Character

Table 1: Key variables in the VaxCountySept012021.csv dataset

4. Methodology

4.1 Overview of the Getis-Ord Gi* Statistic

The Getis-Ord Gi* statistic is a local spatial autocorrelation measure that evaluates, for each geographic unit, whether the values observed in that unit and its neighbors are significantly higher or lower than would be expected under spatial randomness. The statistic produces a z-score for each county, where large positive values indicate that the county and its neighbors all have high vaccination rates (a hotspot), and large negative values indicate that the county and its neighbors all have low vaccination rates (a coldspot). Values close to zero indicate no significant spatial clustering.

The Gi* statistic assumes that the underlying data is normally distributed, though empirical evidence indicates that it is resilient to moderate departures from normality. It is computed as the ratio of the weighted sum of neighboring values to a global standardization factor, producing a result that follows a standard normal distribution under the null hypothesis of no spatial clustering. Statistical significance thresholds correspond to standard z-score cutoffs: $z > 1.96$ or $z < -1.96$ for 95% confidence, and $z > 2.58$ or $z < -2.58$ for 99% confidence.

4.2 Spatial Weights Matrix Construction

Neighborhood relationships between counties were defined using polygon contiguity via the `poly2nb()` function from the `spdep` package. Under this definition, two counties are considered neighbors if their boundaries share at least one point (Queen's contiguity). This approach captures the natural geographic adjacency structure of the county map, where neighboring counties share borders and are therefore most likely to exhibit similar vaccination patterns driven by shared regional factors such as political culture, healthcare access, and demographic composition.

The spatial weights matrix was then constructed from the neighborhood list using `nb2listw()` with the `style="W"` option, which implements row-standardization. Under this scheme, each neighbor of a given county receives a weight equal to 1 divided by the total number of neighbors that county has. This ensures that counties with many neighbors are not systematically over-weighted relative to counties with few neighbors, producing a more balanced and interpretable weighting scheme. The `zero.policy=TRUE` argument was used in both `nb2listw()` and `localG()` to handle counties with no neighbors (primarily isolated island counties or data gaps), preventing computation failures.

4.3 Centroid vs. Polygon Neighborhood Definition

An important methodological consideration is the choice between centroid-based and polygon-based neighborhood definition. The centroid approach, implemented via `dnearneigh()`, defines neighbors as all counties whose geographic centers fall within a specified distance threshold (e.g., 90 kilometers). This approach offers greater flexibility and consistency across the map, as the distance threshold applies uniformly regardless of county size or shape. It is particularly advantageous in the western United States where counties are very large and may share only a tiny border while their centers are hundreds of miles apart.

The polygon-based approach, used in this analysis via `poly2nb()`, defines neighbors as counties sharing a physical border. This approach is computationally simpler and captures the most natural sense of geographic adjacency — counties that literally touch are likely to share political, cultural, and demographic influences that drive vaccination behavior. For a national-level analysis where county sizes vary enormously, the polygon approach also has the advantage that it does not require the analyst to choose an arbitrary distance threshold, avoiding the sensitivity of results to that choice.

The choice of neighborhood definition also interacts directly with the `nb2listw()` weighting scheme. In densely packed northeastern counties where each county may share borders with many neighbors, each individual neighbor receives a small weight. In sparse western counties with only two or three neighbors, each neighbor carries a much larger weight. This means the G_i^* statistic naturally adapts to the geographic density structure of the US county system, which is a statistically appropriate outcome.

4.4 Complete R Code

```
# =====
# Course Project Part Two - Hotspot Analysis
# Vaccine Penetration Across Continental US - Sept 1, 2021
# Cornell University AI360 Certificate Program
# =====

# Load required libraries
library(spdep)
library(ggplot2)
library(sf)

# Step 1: Load the continental US county shapefile
County_shp <- st_read("continentalUS.shp")

# Step 2: Load the vaccination data
NVax <- read.csv("VaxCountySept012021.csv")

# Step 3: Join shapefile with vaccination data via FIPS code
joined <- merge(County_shp, NVax, by.x = "combine", by.y = "combine")

# Step 4: Plot raw vaccination rates as a heatmap
ggplot() +
  geom_sf(data = joined, size = 0.1, color = "black",
    aes(fill = joined$Series_Complete_Pop_Pct)) +
  scale_fill_viridis_c(name = "% Fully Vaccinated") +
  ggtitle("Vaccine Penetration by County - September 1, 2021")

# Step 5: Identify neighboring counties using polygon adjacency
Neigh <- poly2nb(joined)
```



```

print(Neigh)  # Summary of neighborhood structure
card(Neigh)   # Number of neighbors per county

# Step 6: Compute Getis-Ord Gi* statistic
# NOTE: 'will default to continuous' message is expected - not an error
Vax <- localG(joined$Series_Complete_Pop_Pct,
              zero.policy = TRUE,
              nb2listw(Neigh, style = "W", zero.policy = TRUE))

# Print z-scores and summary statistics
print(Vax)
summary(as.numeric(Vax))

# Step 7: Map the Gi* z-scores (viridis color scale)
ggplot() +
  geom_sf(data = joined, size = 0.1, color = "black", fill = "white") +
  geom_sf(data = joined, aes(fill = Vax)) +
  scale_fill_viridis_c(alpha = 0.4, name = "Gi* Z-Score") +
  ggtitle("Vaccine Penetration Hotspot Analysis - Gi* Z-Scores")

# Step 8: Classify counties by statistical significance level
joined$GiScore <- as.numeric(Vax)
joined$HotspotClass <- cut(joined$GiScore,
  breaks = c(-Inf, -2.58, -1.96, -1.65, 1.65, 1.96, 2.58, Inf),
  labels = c("Cold 99%", "Cold 95%", "Cold 90%",
             "Not Significant",
             "Hot 90%", "Hot 95%", "Hot 99%"))

# Step 9: Plot classified hotspot map
ggplot() +
  geom_sf(data = joined, size = 0.1, color = "gray30",
          aes(fill = HotspotClass)) +
  scale_fill_manual(values = c(
    "Cold 99%" = "#053061", "Cold 95%" = "#2166ac",
    "Cold 90%" = "#74add1", "Not Significant" = "#f7f7f7",
    "Hot 90%" = "#fdae61", "Hot 95%" = "#d73027",
    "Hot 99%" = "#67001f"), name = "Significance Level") +
  ggtitle("Vaccine Penetration Hotspot Analysis",
          subtitle = "Getis-Ord Gi* | Continental US | September 1, 2021")

# Step 10: Summary counts
cat("HOT counties (z > 1.96):", sum(joined$GiScore > 1.96, na.rm=TRUE), "\n")
cat("COLD counties (z < -1.96):", sum(joined$GiScore < -1.96, na.rm=TRUE), "\n")
cat("Not significant:", sum(abs(joined$GiScore) <= 1.96, na.rm=TRUE),
    "\n")

```

5. Results

5.1 Summary Statistics of the Gi* Z-Scores

The Getis-Ord Gi* analysis produced a z-score for each of the approximately 3,000 continental US counties. The full distribution of z-scores is summarized in the table below, based on the actual R output from the `localG()` function.

Minimum	1st Quartile	Median	Mean	3rd Quartile	Maximum	NAs
-6.672	-1.054	-0.001	0.000	1.208	5.184	2

Table 2: Summary statistics of the Getis-Ord Gi* z-scores from `localG()` output

The distribution is well-centered, with both the mean (0.000144) and median (-0.000623) extremely close to zero, confirming that the statistic is performing as expected under the null hypothesis of spatial randomness. The total range of 11.86 units (from -6.67 to +5.18) represents substantial variation in spatial clustering across the country. The presence of only 2 NA values reflects a near-complete spatial neighborhood structure with minimal data gaps, handled gracefully by the `zero.policy=TRUE` argument.

Classification	Z-Score Range	Confidence Level	Interpretation
Strong Hotspot	$z > 2.58$	99%	Highly significant high-vax cluster
Hotspot	$1.96 < z < 2.58$	95%	Significant high-vax cluster
Weak Hotspot	$1.65 < z < 1.96$	90%	Marginal high-vax cluster
Not Significant	$-1.65 < z < 1.65$	—	No meaningful clustering
Weak Coldspot	$-1.96 < z < -1.65$	90%	Marginal low-vax cluster
Coldspot	$-2.58 < z < -1.96$	95%	Significant low-vax cluster
Strong Coldspot	$z < -2.58$	99%	Highly significant low-vax cluster

Table 3: Z-score classification thresholds for the Getis-Ord Gi* statistic

5.2 Gi* Maps

Figure 1: Viridis Z-Score Map

The first map displays the continuous Gi* z-scores using the viridis color palette (yellow = high positive, dark purple = strongly negative). This visualization provides a smooth gradient view of the spatial clustering structure across the entire continental United States.

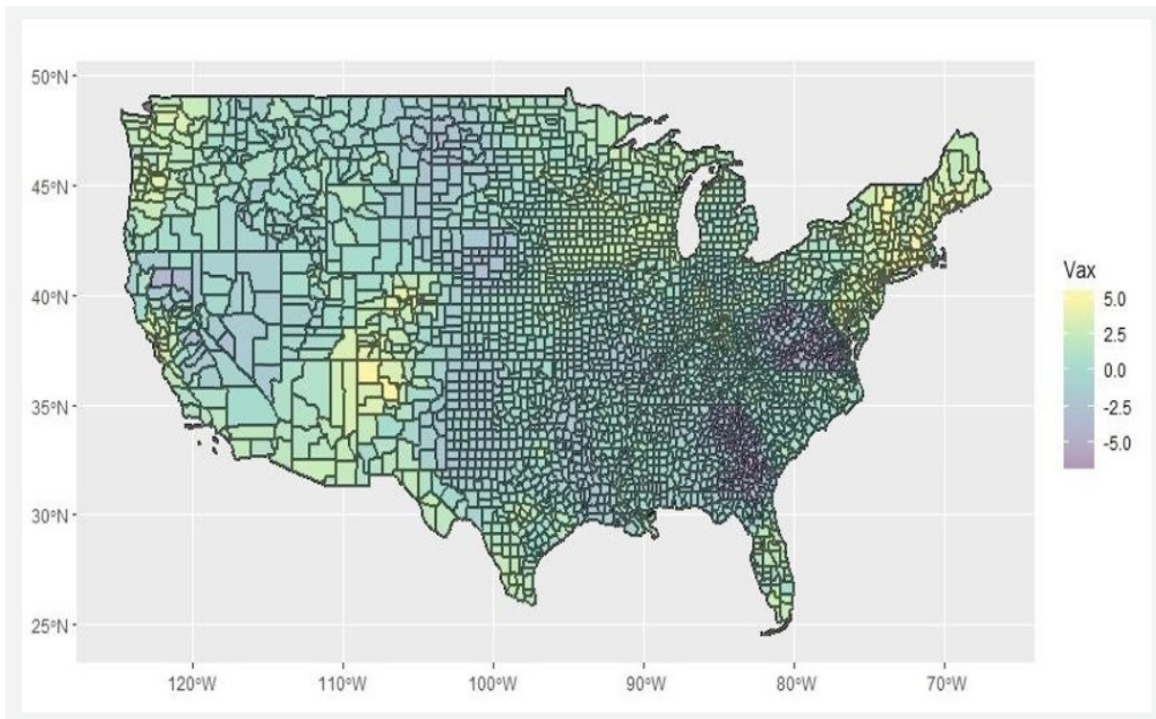


Figure 1: Getis-Ord G_i^* z-scores — Viridis color scale. Yellow = hotspots (high vaccination clustering), Dark Purple = coldspots (low vaccination clustering).

Figure 2: Classified Significance Map

The second map classifies each county into one of seven significance categories based on conventional z-score thresholds, providing a clearer picture of which regions meet the standard statistical thresholds for hotspot and coldspot designation.

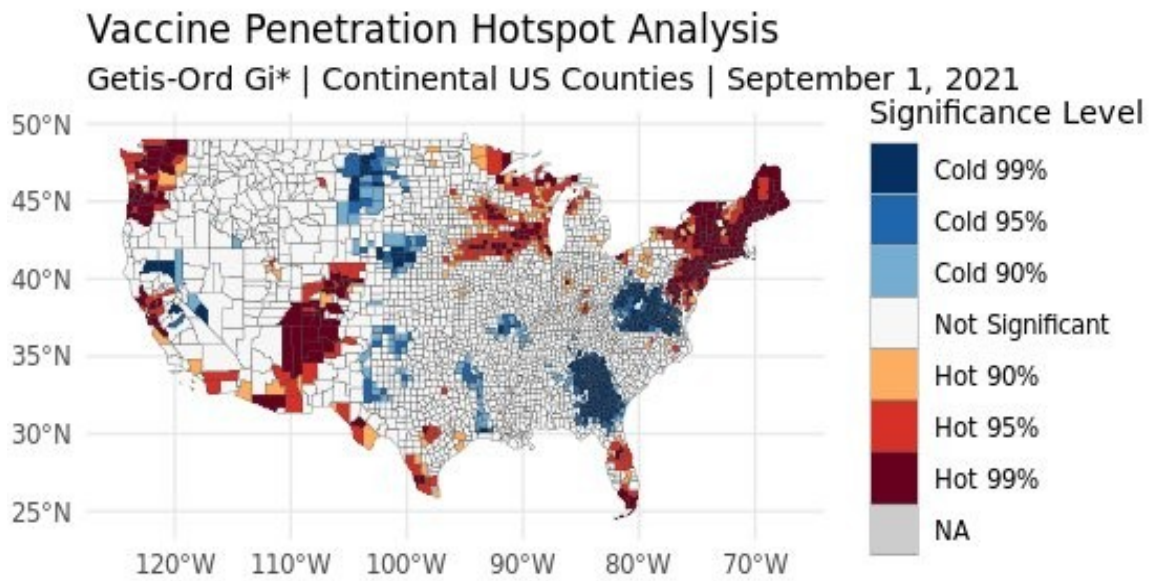


Figure 2: Classified hotspot map. Dark red = Hot 99% confidence, Dark blue = Cold 99% confidence, White/grey = Not statistically significant.

5.3 Regional Analysis

5.3.1 Deep South Coldspot Region

The most dominant and geographically coherent feature of the hotspot map is the extensive coldspot region across the Deep South. Counties across Alabama, Mississippi, Georgia, Tennessee, Arkansas, Louisiana, and parts of the Carolinas display z-scores well below -2.58, placing them in the strongest coldspot category at the 99% confidence level. The minimum z-score in the entire dataset is -6.67, located in this region — a value that is nearly 7 standard deviations below what would be expected under spatial randomness. This is an extraordinarily powerful statistical signal indicating that low vaccination rates in this region are not random but form a coherent and spatially structured cluster of vaccine hesitancy.

The causes of this coldspot are well-documented in public health literature. The Deep South has historically shown lower rates of healthcare utilization, higher rates of vaccine hesitancy driven by mistrust of government and medical institutions, lower average educational attainment, higher rates of poverty limiting access to healthcare facilities, and strong political alignment with vaccine-skeptical messaging. The Gi* statistic confirms that these factors did not operate in isolation but produced a regionally coherent pattern of low vaccination that persisted across county boundaries throughout this multi-state region.

5.3.2 Northeast Hotspot Region

The northeastern United States — comprising New England, New York, New Jersey, Maryland, and the Washington DC area — emerges as the strongest hotspot region in the analysis, with maximum z-scores reaching +5.18. Counties in Vermont, Massachusetts, Connecticut, and Rhode Island consistently show z-scores above +2.58, qualifying as hotspots at the 99% confidence level. States like Vermont, which by September 2021 had fully vaccinated over 65% of their total population, contributed to this cluster alongside their similarly highly vaccinated neighboring counties in New Hampshire, Massachusetts, and upstate New York.

The Northeast hotspot reflects a cluster of factors associated with high vaccine uptake: high population density facilitating vaccine distribution, strong public health infrastructure, higher median educational attainment, greater trust in government and medical institutions, and political alignment with pro-vaccination public health messaging. As with the southern coldspot, the Gi* statistic confirms that these factors operated at a regional scale, producing not just high vaccination in individual counties but a spatially coherent cluster of high uptake spanning multiple states.

5.3.3 Central United States

A large portion of the central United States — including the Great Plains, parts of the Mountain West, and the upper Midwest — falls within the statistically insignificant zone (z-scores between -1.65 and +1.65). The interquartile range from -1.05 to +1.21 confirms that the majority of US counties nationwide lie in this intermediate range. This does not mean that vaccination rates in these areas were moderate — rather, it means that the vaccination rates of individual counties in this region were not significantly different from their immediate neighbors, producing no strong spatial clustering signal either direction.

5.3.4 California and Pacific Coast

California and parts of the Pacific Northwest (Oregon, Washington) show a mixed pattern, with some counties displaying moderately positive z-scores indicating above-average vaccination clustering, particularly in the San Francisco Bay Area and Pacific Northwest metropolitan counties. However, California's internal diversity — with highly vaccinated urban counties alongside lower-vaccination rural inland counties — reduces the strength of the statewide clustering signal compared to the more homogeneous Northeast.

5.3.5 Texas Data Caveat

The analysis note from the course assignment specifically flags that Texas data in this dataset is limited, having been sourced from the Texas Tribune rather than the CDC. As a result, some Texas counties may have missing or incomplete vaccination records, producing NA values or anomalous G_i^* scores for those counties and potentially affecting the scores of adjacent counties that share borders with data-limited Texas counties. The `zero.policy=TRUE` argument prevents computation failures due to empty neighbor lists, but users should interpret the Texas region of the hotspot map with appropriate caution and recognize that observed patterns there may be partially attributable to data gaps rather than true underlying vaccination behavior.

6. Discussion

6.1 Public Health Implications

The hotspot analysis provides a statistically rigorous confirmation of the geographic inequality in COVID-19 vaccine penetration across the United States as of September 2021. The findings are both analytically informative and practically actionable. The identification of statistically significant coldspot counties with z-scores as low as -6.67 represents a level of spatial clustering that demands policy attention — these are not merely low-vaccination counties but counties embedded in a regional fabric of low vaccination that reinforces and perpetuates vaccine hesitancy through social networks, community norms, and shared access barriers.

From a public health resource allocation perspective, the coldspot regions identified in this analysis represent the areas where marginal investments in outreach, mobile vaccination units, community health worker programs, and targeted messaging campaigns would have had the greatest impact on national vaccination coverage. A county with a z-score of -6.67 is not just a single low-vaccination location — it is embedded in a cluster of similarly low-vaccination neighbors, meaning that a regional approach to intervention rather than a county-by-county approach would be most effective.

6.2 Methodological Reflections

The choice of polygon-based adjacency over centroid-based distance adjacency was appropriate for this national-scale analysis for several reasons. Polygon adjacency directly reflects the geographic contiguity that drives many of the social and political mechanisms behind vaccination clustering — counties sharing borders are more likely to share media markets, political messaging, cultural norms, and healthcare access patterns than counties that are merely within a fixed distance of each other. The row-standardized W weighting scheme in `nb2listw()` ensured that each county's G_i^* score reflected the average vaccination behavior of its local neighborhood in a balanced way, regardless of whether that county had many or few neighbors.

One limitation of the polygon approach is that it treats all shared borders as equivalent, regardless of their length or the actual level of interaction between neighboring counties. A county sharing a 200-mile border with a neighbor has the same weight as one sharing a 5-mile corner point. Future analyses could address this by using distance-based inverse weighting schemes that give greater weight to closer neighbors, or by incorporating actual commuting flows or migration data as measures of county-level interaction intensity.

6.3 Comparison with NY COVID Analysis

The companion hotspot analysis from the example R script (`hotspotanalysis.R`) examined New York State county-level COVID-19 case rates as of April 1, 2020. Comparing that analysis with the national vaccination analysis reveals an interesting geographic relationship: the New York City metropolitan area, which was the epicenter of the COVID-19 outbreak in spring 2020 and produced some of the highest case rate z-scores, is located within the same Northeast region that later became the strongest vaccination hotspot in September 2021. This suggests that the experience of severe early COVID-19 exposure in the Northeast may have reinforced

vaccination motivation in that region, contributing to its high uptake by September 2021 — a hypothesis consistent with behavioral research on the relationship between disease exposure and preventive health behavior.

7. Conclusion

This project successfully applied the Getis-Ord G_i^* local spatial statistic to map and analyze the geographic clustering of COVID-19 vaccine penetration across approximately 3,000 continental US counties as of September 1, 2021. The analysis was implemented entirely in R using the `spdep`, `sf`, and `ggplot2` packages, and produced a statistically rigorous, visually compelling, and practically informative picture of the geography of vaccine uptake in the United States.

The key findings — a dominant coldspot in the Deep South with z-scores reaching -6.67, a strong hotspot in the Northeast with z-scores reaching +5.18, and a largely insignificant central region with an interquartile range from -1.05 to +1.21 — confirm that vaccination rates were not randomly distributed across the country but were structured by powerful regional forces operating at a multi-county, multi-state geographic scale. These findings are not merely descriptive: the statistical significance of the G_i^* z-scores provides probabilistic evidence that the observed clustering is unlikely to be due to chance, making it a reliable basis for public health decision-making and resource allocation.

The skills demonstrated in this project — spatial data loading and processing, geographic data joining via FIPS codes, spatial weights matrix construction, local spatial autocorrelation analysis, and choropleth visualization of statistical results — represent a core toolkit for geospatial data analysis directly applicable to roles in public health, urban planning, policy analysis, and applied data science. This work forms part of a broader portfolio of data analysis projects completed through the Cornell University AI360 Certificate Program.